

Abstract

Project Title – IOT based- Life Health Band -Health Monitoring System for Elderly People.

Type – In- House

Introduction –

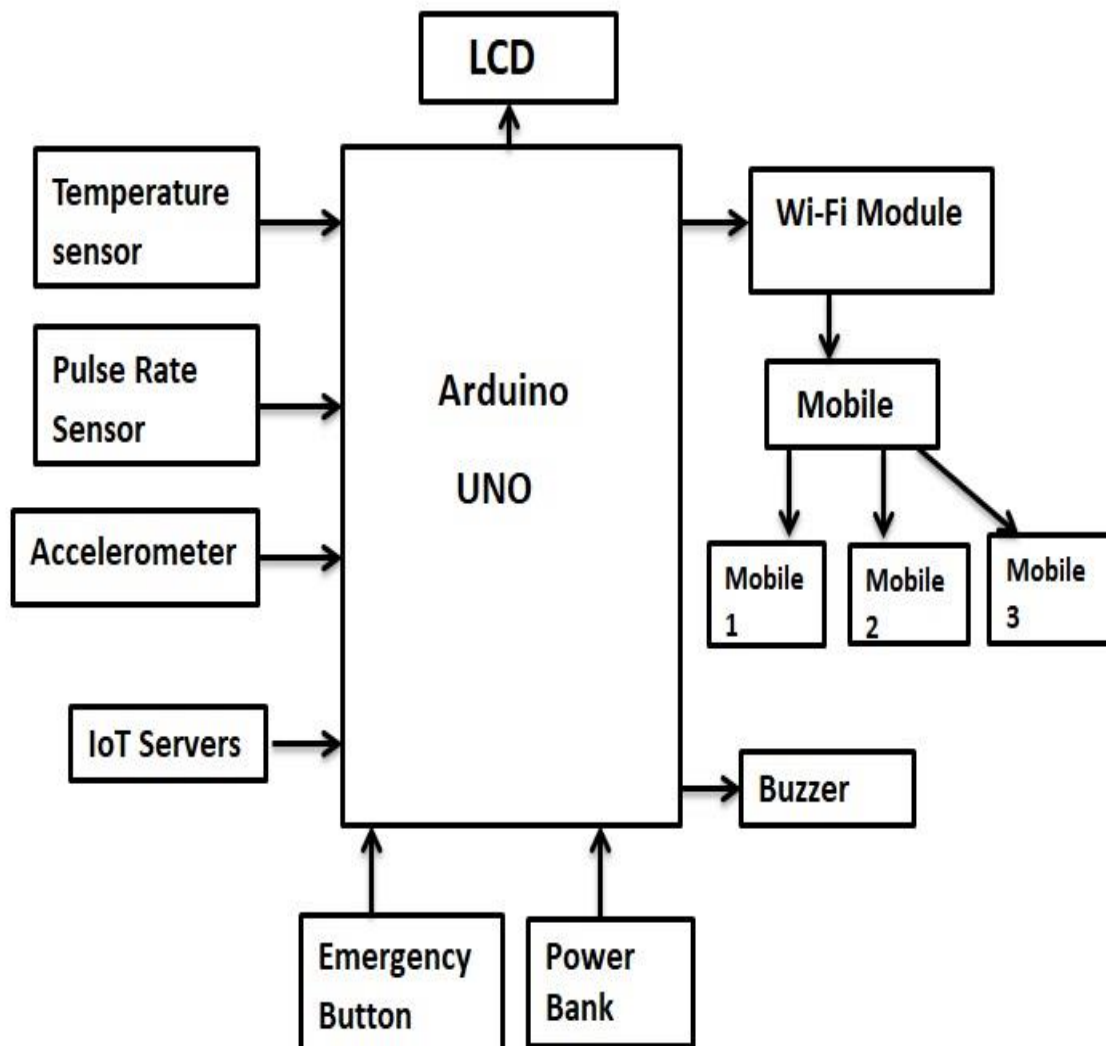
Old aged peoples, Alzheimer's affected ones, persons with heart problems always need an attention. If there is a situation that no one is present at home, during these times no one will be there to take care of them. Even if they fall down, a mild stroke occurs, temperature increases the first aid will only get when someone notice it, it may take long time since no one is there. This may lead to situations that we lose them.

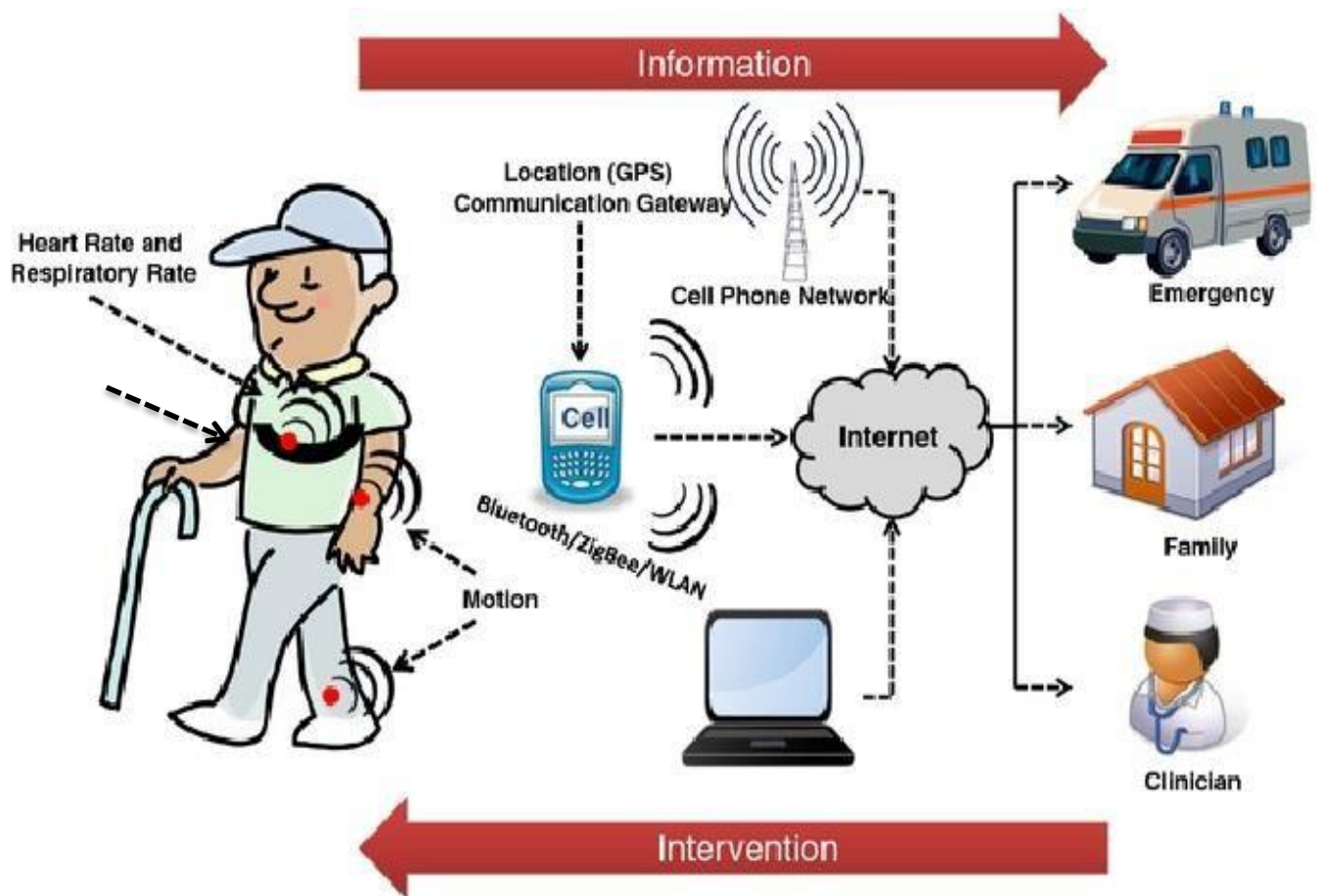
Health Band is an innovative solution for detecting a person whose health in in grave danger. It's mostly designed as a remote rescue system for people who are at risk of having stroke, cardiac arrest and heart attack. Health Band is a smartphone synced mobile health monitoring bracelet capable of reading a human's vital signs (pulse rate and body-temperature)

The Internet of Things (IoT) has been widely used to interconnect the available medical resources and offer smart, reliable, and effective healthcare service to the elderly people. Health monitoring for active and assisted living is one of the paradigms that can use the IoT advantages to improve the elderly lifestyle. In this project, we present an IoT architecture customized for healthcare applications. The proposed architecture collects the data and relays it to the cloud where it is processed and analysed. Feedback actions based on the analyzed data can be sent back to the user.

Thus by the smart Heath band we can take care of our beloved ones from any were in the world.

Block Diagram –





Working –

We propose a portable and customizable IoT system that can be used to collect the data needed to facilitate the independent living of senior and challenged citizens to improve their quality of life. The proposed system design philosophy targets are having a low-power system that can be worn during the day and be turbo charged as necessary. The system is designed to be light and comfortable to wear. Furthermore, the system is implemented using low-cost components, which makes it an affordable system

In addition to the above medical sensor that capture the user's health signs, light and fall detection sensors are also used. The fall detection sensor is an accelerometer that is used to determine whether the user has fallen abruptly, Temperature changes, continuous pulse rate counting and changes in pulse rate such that warnings signals can be generated to provide the needed help.

Controller is responsible for collecting the data of the different sensors interfaced to it, and communicating such a data to the server for further processing. The Wi-Fi transceiver is used to connect the proposed system to the Internet. The collected user's data is communicated to a server which is responsible for facilitating the accessibility of such a data anywhere through the Internet.

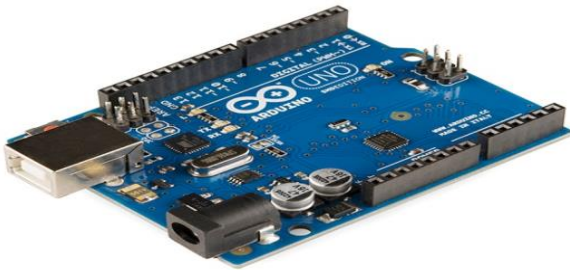
The data can be accessed and manipulated through software tools. This way the patient is continuously monitored by system ,and if there are some changes in expected readings occurs then warning will sent automatically to the users doctor, family as a feedback through IoT network.

Specifications –

1. Arduino UNO- 8 bit ATmega328P (Freq-16MHZ)
2. Pulse Sensor(SEN-11574) - 0.625" Diameter and 0.125" Thick
3. Temperature Sensor – LM35
4. Accelerometer - ADXL335 Triple Axis , 6 pin Accelerometer
5. Emergency button- 6x6x5mm Tactile Push Button
6. 16x2 LCD Display
7. WiFi Module ESP8266
8. Android/IOS Mobile
9. Power Bank/9V battery
10. Buzzer

Hardware and Software –

1. Arduino UNO- The **Arduino Uno** is an Open source microcontroller board based on the Microchip ATmega328P microcontroller. The board is equipped with sets of digital and analog (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable. It can be powered by the USB cable or by an external 9 volts, though it accepts voltages between 7 and 20 volts. It is similar to the Arduino Nano and Leonardo. The hardware reference design is distributed under a Creative Commons Attribution Share-Alike 2.5 license and is available on the Arduino website. Layout and production files for some versions of the hardware are also available.



Technical Specification –

- Operating Voltage: 5 Volts
- Input Voltage: 7 to 20 Volts
- Digital I/O Pins: 14 (of which 6 can provide PWM output)
- UART: 1
- I2C: 1
- SPPI: 1
- Analog Input Pins: 6
- DC Current per I/O Pin: 20 mA
- DC Current for 3.3V Pin: 50 mA
- Flash Memory: 32 KB of which 0.5 KB used by bootloader
- SRAM: 2 KB
- EEPROM: 1 KB
- Clock Speed: 16 MHz
- Length: 68.6 mm
- Width: 53.4 mm
- Weight: 25 g

2. Accelerometer – An accelerometer is an electromechanical device that will measure acceleration forces. These forces may be static, like the constant force of gravity pulling at your feet, or they could be dynamic - caused by moving or vibrating the accelerometer.

By measuring the amount of static acceleration due to gravity, you can find out the angle the device is tilted at with respect to the earth. By sensing the amount of dynamic acceleration, you can analyze the way the device is moving.

Accelerometer will be used to locate the vip person on the x, y, and z axis parameters. The accelerometer will provide the location of the vip within the parliament location.



Technical specification –

- 3-axis sensing.
- Small, low profile package.
- 4mm × 4 mm × 1.45 mm LFCSP.
- Low power : 350 μ A (typical)
- Single-supply operation: 1.8 V to 3.6 V.
- 10,000 g shock survival.
- Excellent temperature stability.
- BW adjustment with a single capacitor per axis.

3. Temperature sensor: LM35 – Temperature sensor is used to sense the temperature. We have used a Temperature sensor called LM35. This temperature sensor can sense the temperature of the atmosphere around it or the temperature of any machine to which it is connected or even can give the temperature of the human body in case if used. So, irrespective of the application to which it is used, it gives the reading of the temperature. The LM35 series are precision integrated-circuit temperature sensors, whose output voltage is linearly proportional to the Celsius (Centigrade) temperature.

Temperature sensor is an analog sensor and gives the output into form of analog signal. This signal is feed to ADC which will convert it into digital form. Once converted into analog form, the microcontroller can process the digital temperature signal as per the application.



Technical Specification –

- Calibrated Directly in Celsius (Centigrade)
- Linear + 10-mV/°C Scale Factor
- 0.5°C Ensured Accuracy (at 25°C)
- Rated for Full –55°C to 150°C Range
- Suitable for Remote Applications
- Low-Cost Due to Wafer-Level Trimming
- Operates From 4 V to 30 V
- Less Than 60-μA Current Drain
- Low Self-Heating, 0.08°C in Still Air
- Non-Linearity Only $\pm\frac{1}{4}^{\circ}\text{C}$ Typical
- Low-Impedance Output, 0.1 Ω for 1-mA Load

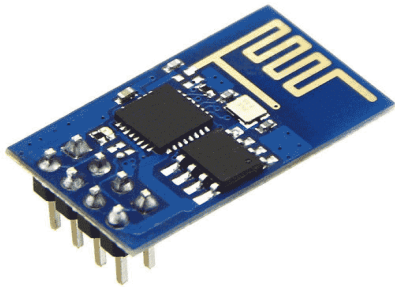
4. Pulse Rate sensor – The Pulse sensor SEN-11574 is type of plug and play sensor most suitable for Arduino. The sensor is very easy to interface and install in any project and gives live Heart rate data. The Module is integrated with the optical amplifier and noise elimination ckt.



Technical Specification –

- Biometric Pulse Rate or Heart Rate detecting sensor
- Plug and Play type sensor
- Operating Voltage: +5V or +3.3V
- Current Consumption: 4mA
- Inbuilt Amplification and Noise cancellation circuit.
- Diameter: 0.625"
- Thickness: 0.125" Thick

5. Wi-Fi module for Arduino – The **Arduino Uno Wi-Fi** is an **Arduino Uno** with an integrated **Wi-Fi module**. The board is based on the ATmega328P with an **ESP8266WiFi Module** integrated. The **ESP8266WiFi Module** is a self-contained SoC with integrated TCP/IP protocol stack that can give access to your **Wi-Fi** network (or the device can act as an access point)



Technical Specification –

- 2.4 GHz Wi-Fi (802.11 b/g/n, supporting WPA/WPA2).
- General-purpose input/output (16 GPIO).
- Inter-Integrated Circuit (I²C) serial communication protocol.
- Analog-to-digital conversion (10-bit ADC).
- Serial Peripheral Interface (SPI) serial communication protocol.
- I²S (Inter-IC Sound) interfaces with DMA (Direct Memory Access) (sharing pins with GPIO).
- UART (on dedicated pins, plus a transmit-only UART can be enabled on GPIO2).
- Pulse-width modulation (PWM).

Fig : Circuit Simulation Diagram

Results –

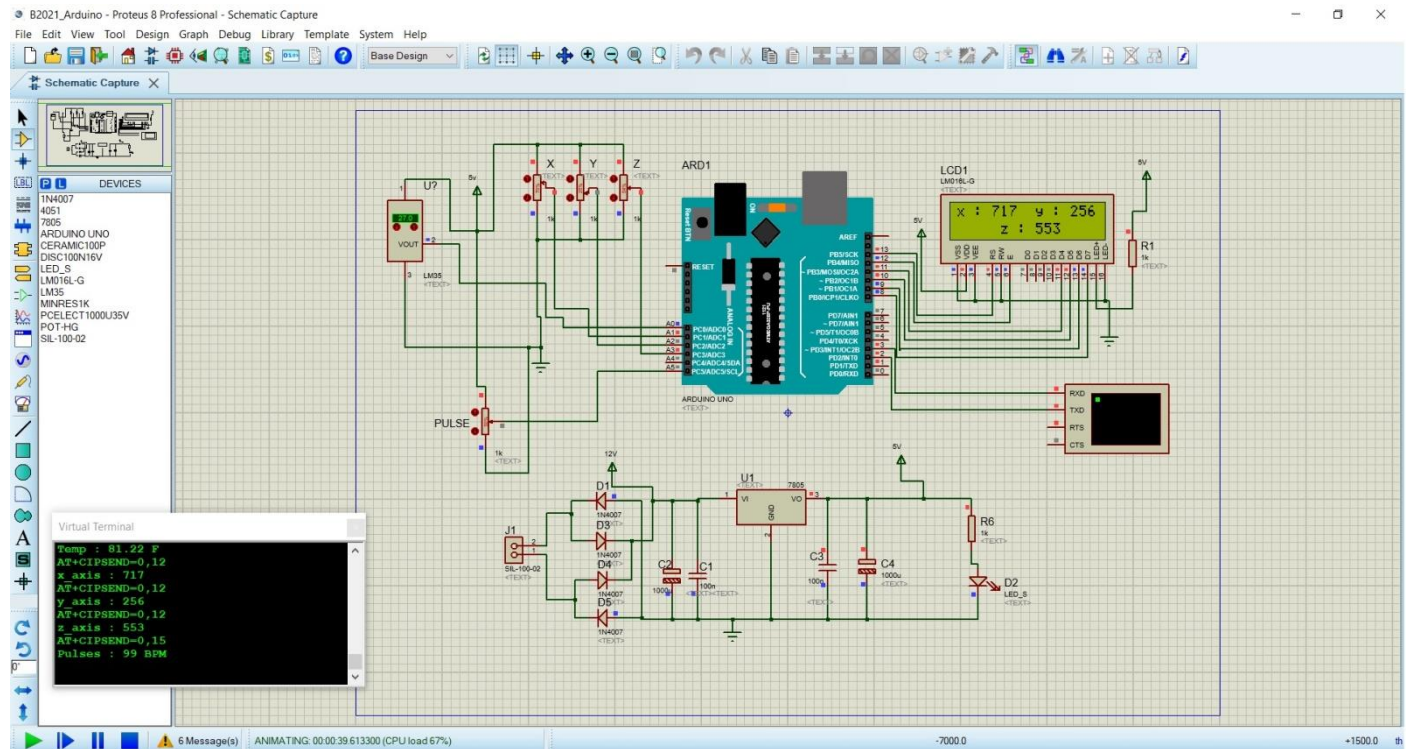


Fig : Simulation Accelerometer Output

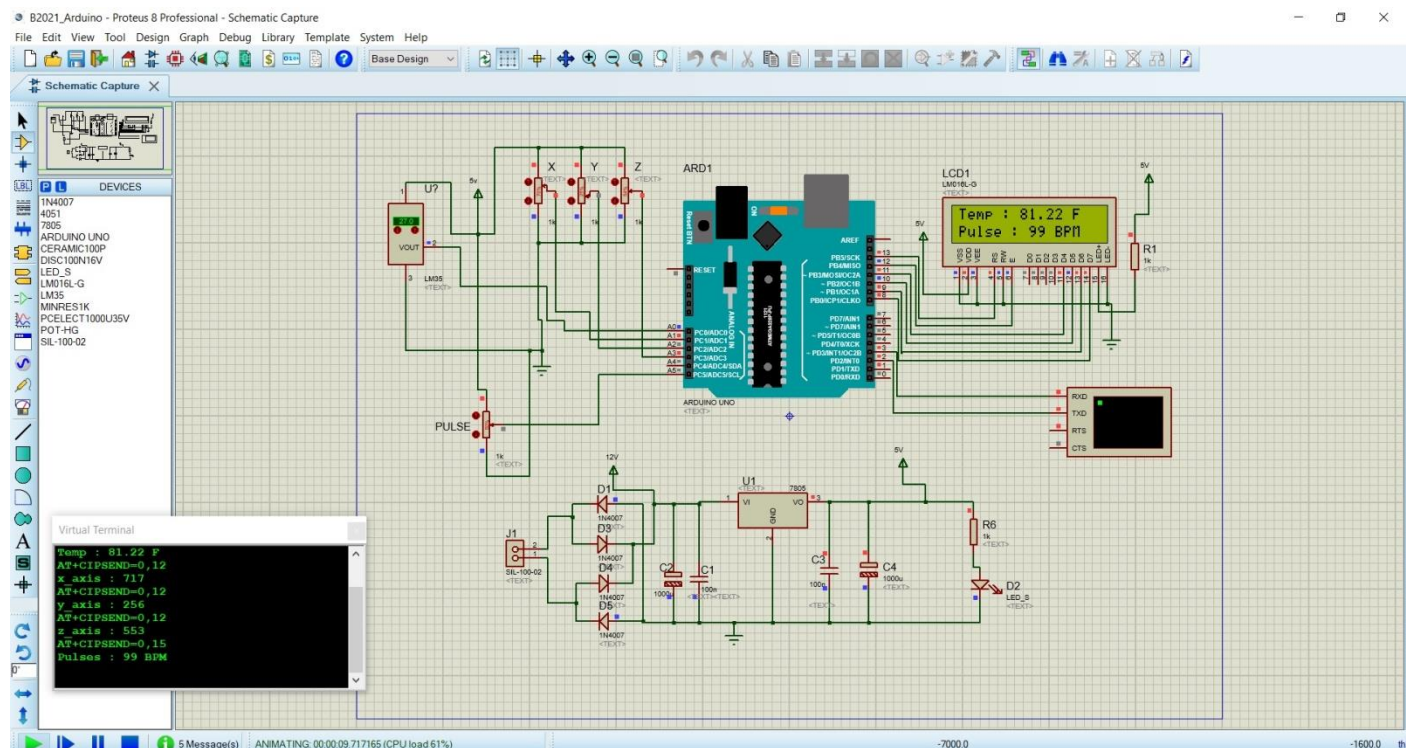


Fig : Simulation Temperature and Pulse rate Sensor Output

Software Requirements –

- **Arduino Software IDE:**

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

- Inexpensive
- Cross-platform
- Simple, clear programming environment
- Open source and extensible software
- Open source and extensible hardware.

- Make circuit in proteus, PCB making, Soldering, H/W testing, Software coding on Embedded C using KEIL Compiler.
- To design an IOT SERVER software in Fire Fox. To design a GUI containing Tables and Graphs.

- **MQTT for IOT based Applications.**

- MQTT stands for Message Queuing Telemetry Transport.
- It's a simple publish-and-subscribe system that allows you to send and receive messages as a client.



- MQTT is a low-bandwidth messaging protocol optimized for limited applications. As a result, it's ideal for Internet of Things applications.
- You can use MQTT to monitor outputs, read and publish data from sensor nodes, and much more.
- As a result, establishing contact between multiple devices is incredibly simple.

Advantages-

- Health Monitoring System provides real-time analysis of the patient's health parameters. Patients are sharing their healthcare information in real time with their caregivers through these devices for flexible health monitoring and management.
- Internet of medical things is vastly improving the healthcare condition of patients.
- It allows patients to maintain independence, prevent complications.
- In addition, patients and their family members feel comfortable knowing that they are being monitored and will be supported if a problem arises.
- Reduce burden for caretakers.
- Improve confidence on their wellbeing.

Applications –

- Monitor the motions and status of the person, heart beat rate, temperature, up-to-date.
- A Sudden fall due to heart attacks or losing balance, or abnormal fall or rise in heart beat rate it alerts the nearby ones of the person to get immediate attention
- The Health band also measures the temperature of the person in order to understand the person health condition.

References:

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- **SparkFun Triple Axis Accelerometer Breakout - ADXL335 - SEN ...**
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 - SEN-09269: Breakout board for the 3 axis ADXL335 from Analog Devices.
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 - www.analog.com/en/products/mems/accelerometers/adxl335.html
- The ADXL335 is a small, thin, low power, complete 3-axis accelerometer
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- *lcdsmartie.sourceforge.net*
- www.8052.com/tutlcd2.phtml

Group Details						
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